Project Specification Document: Sports Club Management Website

1. Project Overview

The goal of this project is to design and develop a comprehensive sports club management website that enables seamless interaction between athletes, coaches, and administrators. The platform will streamline registration, profile management, competition participation, and fee tracking—similar in functionality to any club or association in for state or national games

Objectives

- Provide a centralized platform for athletes and coaches to register and manage profiles.
- Simplify competition registration and fee payments.
- Enable administrators to efficiently manage competitions, participants, and payments.
- Maintain a structured database of athletes with verified documents.
- Improve transparency with real-time activity tracking and reporting.

Target Users

- **Athletes** (Students / Participants)
- **Coaches / Trainers**
- **Club Administrators**

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2. Core Platform Features

2.1. User Registration & Authentication

- Athlete registration with detailed personal information.
- Coach registration with credentials and experience.
- Secure login system for all users.
- OTP/email verification for account activation.

2.2. Athlete Profile Management

- Create and edit athlete profiles.
- Track participation history.

2.3. Coach Profile Management

- Create coach profiles with qualifications, experience, and certifications.

- Ability to manage assigned athletes (optional future features).

2.4. Competition Management

- Admin can create and manage competitions categorized by **Age Group** and **Skill level**.
- Athletes can view and register for relevant competitions.
- Automatic filtering of competitions based on the athlete's profile (age/skill).

2.5. Fee Payment System

- Online payment integration (UPI, Credit/Debit Cards, Net Banking).
- Payment gateway required for competition participation.
- Auto-generated payment confirmation and receipts.
- Payment status tracking (Paid / Pending / Failed).

2.6. Admin Dashboard

A powerful dashboard with real-time insights:

- **Metrics:** Total registered athletes, participants per competition.
- **Activity Feed:** New registrations, recent payments.
- **Payment Tracking:** Lists of users who have paid vs. users with due payments.
- **Analytics:** Competition-wise analytics and reporting.

2.7. Data Management System

- Centralized database of all athletes with access to full registration forms.
- Document verification system (Admin mark: Approved / Rejected / Pending).
- Search and filter options (By name, age group, competition).
- Data Export functionality (Excel/PDF).

2.8. Notifications & Alerts

- Payment reminders for due fees.
- Competition updates and registration confirmation messages via Email/SMS.

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3. Athlete Registration Form Specification (Developer Blueprint)

This multi-step form is the core onboarding engine for athletes. It must support state-preservation (save & continue) and real-time inline validation.

Step 1: Personal Details

- **Full Name** (as per official documents)
- **Date of Birth (DOB)**
- **Age** *(Auto-calculated based on DOB)*
- **Gender** (Dropdown)
- **Blood Group** (Dropdown)
- **Mobile Number** (Must validate as exactly 10 digits)
- **Email Address** (Must validate standard email format)

Step 2: Parent / Guardian Details

- **Father's Name**
- **Mother's Name**
- **Guardian's Name** (If applicable)
- **Parent/Guardian Mobile Number**
- **Parent/Guardian Email** (Optional)

Step 3: Address Details

- **Current Address** (Full address with landmark)
- **City / District**
- **State** *(Feature: Auto-detect state from PIN code via API)*
- **PIN Code**
- **Country** (Default: India)

Step 4: Club / Representation Details

- **Club Name**
- **State Representation**
- **District**
- **Upload NOC** (No Objection Certificate) from:
    - Club
    - State Association

Step 5: Competition Details

- **Age Group Applied For** *(Auto-suggested/restricted based on DOB)*
- **Category / Level** (Beginner / Intermediate / Advanced)
- **Event / Competition Selection** (Multi-select permitted if applicable)

Step 6: Document Upload Section (Includes Insurance)

All uploads must feature a thumbnail preview and a "Remove" option.

Mandatory Uploads:

- Passport Size Photo
- Aadhaar Card / ID Proof
- Birth Certificate / DOB Proof
- School Bonafide Certificate

Insurance Details (Conditional/Mandatory based on Admin settings):

- Insurance Provider Name
- Policy Number
- Valid Till (Expiry Date) -> *Must validate that date is not in the past.*
- Upload Insurance Document (PDF/Image)

Optional Uploads:

- Previous Achievement Certificates
- Medical Fitness Certificate

Step 7: Declaration & Consent

- Checkbox: "I confirm that all details provided are correct."
- Checkbox: "I agree with the terms and conditions."
- **Parent/Guardian consent** *(Mandatory dynamically if Age < 18).*

Step 8: Payment Section (Final Step)

- Registration Fee / Competition Fee Display.
- Payment Gateway Integration (UPI, Cards, Net Banking).
- Payment Status updates (Paid / Pending).

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4. Technical & File Handling Requirements

4.1. File Upload Rules & Auto-Compression

- **Supported Formats:** Images (JPG/JPEG/PNG), Documents (PDF only).
- **File Size Limits:** Max 1 MB for Images, Max 2 MB for PDFs.
- **Smart Auto-Compression (Frontend):**
  - Use browser-based compression libraries before uploading to the server.
  - Maintain 70–80% visual quality.
  - Resize large images to standard dimensions (e.g., 1024px max width).
  - **UI Trigger:** Display message *"Your image has been optimized for faster upload"* upon successful compression.
- **Storage:** Store files securely via cloud storage (e.g., AWS S3).

4.2. UI/UX Recommendations

- **Multi-Step Form:** Implement a progress bar at the top indicating the current step.
- **Conditional Logic:** Only show the Insurance upload section when required for a specific competition, accompanied by a warning: *"⚠️ Insurance is required to participate in this competition."*
- **Drag & Drop:** Enable drag-and-drop zones for document uploads.
- **Responsive:** Ensure that the entire onboarding flow is highly mobile-friendly, as many users will upload photos directly from their phone cameras.

4.3. Admin Controls (Specific to Registration)

- **Verification:** Admin dashboard must allow toggling document status between Approved / Rejected / Pending.
- **Insurance Tracking:** Dashboard must highlight expired insurance in **red**. Filter options needed for: *Insurance Uploaded*, *Insurance Expired*, *Insurance Missing*.
- **Duplicate Prevention:** Backend must prevent duplicate registrations based on Mobile Number / Email.

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5. Future Scope & Advanced Features

- Mobile app integration (iOS/Android).
- Coach-athlete in-app communication system.
- Performance tracking and ranking system.
- QR-based entry scanning for live competitions.
- Live competition results and public leaderboards.

6. Achievement & Certificate Management System (New Module)

Feature Overview: A dedicated system to track an athlete's historical performance, display past competition details, and allow the automated generation and downloading of participation and winning certificates.

6.1. Athlete Dashboard: "My Competitions & Achievements"

Athletes will have a dedicated tab in their profile to view their history.

- **Past Competitions List:** A chronological list of all completed competitions the athlete participated in.
- **Competition Details Displayed:**
    - Competition Name & Date

    - Category / Event Level (e.g., U-14 Beginners)
    - Result / Standing (e.g., Gold, Silver, Bronze, or "Participant")
- **Download Actions:**
    - **Download Certificate:** A button to download the PDF certificate (dynamically unlocked once the admin publishes the results).
    - **Share Feature (Optional):** Quick button to share the certificate or achievement on social media (WhatsApp, Facebook, etc.).

6.2. Admin Dashboard: Post-Competition Management

Admins need tools to finalize competitions and distribute certificates efficiently.

- **Result Publishing System:** Admin can navigate to a completed competition and update the final standings for all registered athletes.
- **Medal Assignment:** Dropdown next to athlete names to assign: *Gold, Silver, Bronze, or None (Participation only).*
- **Certificate Generation (Two Approaches for Developers):**
    - *Option A (Auto-Generation):* The system uses a predefined digital template and automatically overlays the Athlete's Name, Competition Name, Date, and Result to generate a PDF.
    - *Option B (Manual Upload):* Admin can manually upload a custom PDF certificate file for specific athletes.
- **Bulk Actions:** "Generate and Send Participation Certificates to all attendees."
- **Notifications:** Automated Email/SMS sent to athletes: *"Results are out! Download your certificate from your dashboard."*

6.3. Database Schema Updates (Developer Notes)

To support this feature, the backend database will need an extension to track post-competition data.

- **Table:** **`competition_results`** (or added as columns to the `registrations` table)
    - `registration_id` (Linked to the specific athlete and competition)
    - `attendance_status` (Present / Absent - to ensure only attendees get certificates)
    - `medal_won` (Enum: Gold, Silver, Bronze, None)
    - `certificate_url` (Link to the stored PDF file in AWS S3 / Cloud)
    - `published_at` (Timestamp when the admin released the certificate)

6.4. Technical Execution & Library Recommendations

- **PDF Generation:** Developers should use backend libraries like `Puppeteer` (Node.js) to generate high-quality PDFs from HTML templates, or frontend libraries like `jsPDF` or `pdf-lib`.
- **Storage:** Generated certificates should be saved to secure cloud storage (like AWS S3) to prevent the database from becoming too large, with only the file URL saved in the database.

- **Watermarking/QR Verification (Advanced):** Auto-generate a unique QR code on the certificate that, when scanned, links back to the athlete's verified profile on the club website to prove the certificate is authentic.

7. Comprehensive Email/SMS Notification & Reminder System

Feature Overview: A fully automated, event-driven communication layer that triggers personalized emails (and optionally SMS) based on user actions, upcoming deadlines, and administrative announcements.

7.1. Account & Profile Notifications

- **Welcome Email:** Sent immediately upon successful account creation, detailing login instructions and how to complete their profile.
- **OTP / Account Verification:** Sent during the initial signup process to verify the user's email address.
- **Document Status Alert:** Sent when an Admin approves or rejects an athlete's uploaded documents (e.g., "Action Required: Your ID proof was rejected. Please re-upload.").

7.2. Event Registration & Payment Triggers

- **Registration Confirmation:** Sent immediately after an athlete successfully enrolls in a specific competition, summarizing the event details.
- **Payment Success & Receipt:** Automated email containing the transaction reference, amount paid, and a downloadable PDF invoice/receipt.
- **Payment Failed Alert:** Sent if a transaction drops or fails, providing a direct link to retry the payment securely.

7.3. Automated Reminders (Scheduled Alerts)

- **Upcoming Event Reminder:** Scheduled emails sent 48 hours and 24 hours before the competition date. These must include the venue, reporting time, category, and a checklist of mandatory items to carry.
- **Pending Fee Reminder:** Automated nudges sent 3 days and 1 day before a registration deadline for users who have initiated registration but have a "Pending" payment status.
- **Missing Document Alert:** Automated reminder for athletes who have registered for an event but are missing mandatory uploads (like Insurance or NOC) before the cutoff date.

7.4. Post-Competition Communications

- **Results Published:** An alert sent to all registered attendees the moment the Admin publishes the final leaderboard.

- **Certificate Available:** A congratulatory email notifying the athlete that their participation or winning certificate is generated and ready to download from their dashboard.

7.5. Technical Execution & Developer Notes

- **Email Service Providers (ESPs):** Developers must integrate reliable transactional email APIs such as Klaviyo, Mailchimp, SendGrid, AWS SES, or Mailgun to ensure high inbox deliverability.
- **Data-Driven Segmentation & Triggering**
  Use Klaviyo's advanced segmentation to target users based on behavior, event timelines, or custom properties, ensuring reminders are sent to the right audience at the right time.
- **Automation & Scheduling (Klaviyo Flows)**
  Leverage Klaviyo's built-in **Flows and automation system** to schedule and trigger time-based reminders automatically.
  Set up event-based triggers (such as upcoming dates or user actions) to send emails at predefined intervals without requiring manual cron job management.
- **Dynamic & Personalized Email Templates**
  Utilize Klaviyo's dynamic template builder with variables and conditional logic to create visually branded emails.
  Automatically populate content such as:
  - User Name
  - Event Name
  - Event Date
  - Custom properties

  This ensures highly personalized communication at scale.
- **Audit Logging:** The database must log all sent emails against the user's profile so Admins can verify if a reminder was successfully delivered.